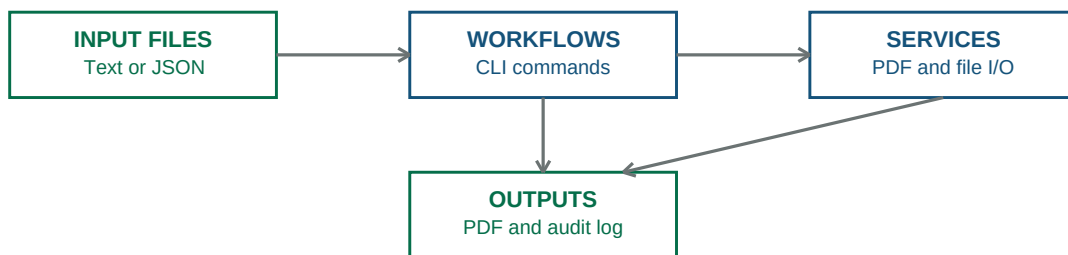


Architecture Overview

Purpose

A local Node.js workflow for producing NABH-aligned Complaint Forms and Admission Advice Notes from structured or plain-text input files.

System Flow



Technology

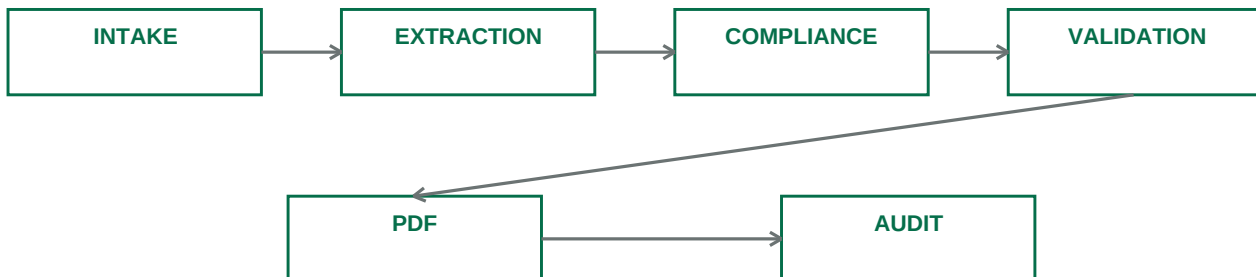
- Node.js runs standalone command-line workflows. The project uses ECMAScript modules via the package type setting.
- pdf-lib is the sole external dependency and renders PDFs directly in JavaScript using standard fonts, text, lines, panels, and checkboxes.
- JSON supplies structured document data; the Complaint workflow can also accept colon-delimited text input.
- The system is file-based. It has no database, HTTP API, authentication layer, or external AI integration.

Project Layout

- workflows/: command-line orchestration for complaint, interactive complaint, admission-advice, and architecture document generation.
- agents/: focused complaint pipeline stages for intake, extraction, compliance, validation, PDF handoff, and audit logging.
- services/: shared file and logging utilities plus independent PDF renderers for each document type.
- templates/, models/, sample/, and inputs/: input contracts, schema, examples, and saved records.

Complaint Processing Pipeline

Complaint Data Pipeline



Pipeline Stages

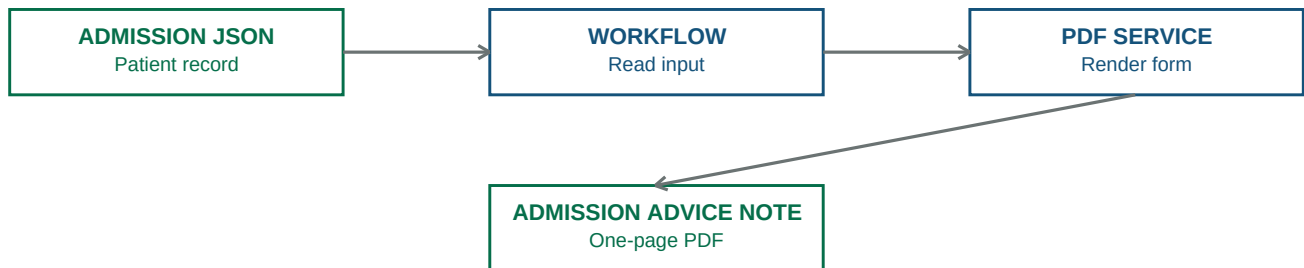
- Intake reads the supplied file as text through the shared file service.
- Extraction accepts JSON directly or maps Label: Value text lines into the complaint record shape.
- Compliance applies complaintSchema.json as the canonical field list. Missing required values become NOT PROVIDED, while selected optional values remain blank.
- Validation returns an isValid flag and the list of fields that remain NOT PROVIDED.
- PDF delegates to the Complaint PDF service, which renders the NABH complaint form.
- Audit writes the complaint ID, patient name, complaint date, and closure status to the console log.

Outputs

- The complete complaint workflow creates Complaint_NABH.pdf and its two existing variant filenames in the project root.
- Individual agents may be run with `npm run agent:<name>` to inspect their intermediate console output.

Admission Advice and Commands

Admission Advice Flow



Admission Advice Flow

- The admission-advice workflow reads an admission JSON record and passes it directly to its PDF service; it does not use the complaint agents.
- The renderer produces a one-page note with patient identification, clinical assessment, routing and ward directives, STAT orders, and financial counseling fields.
- Checkboxes are selected by matching JSON values such as source, priority status, ward, safety precautions, monitoring frequency, diet, and payment pathway.
- Long clinical values are wrapped to fit the designed form layout.

Primary Commands

- `npm run complaint` - process the default complaint sample and generate complaint PDFs.
- `npm run complaint -- ./path/to/complaint.json` - process a specific complaint input file.
- `npm run complaint:interactive` - collect complaint data interactively.
- `npm run admission-advice` - generate a populated admission note from the sample admission JSON.
- `npm run admission-advice -- ./my-admission.json ./My_Admission_Advice.pdf` - generate a named PDF from custom data.
- `npm run architecture` - regenerate this architecture document.